Digital Audio via Unlimited Sensing: Overflow Overcomes Clipping and Overflow

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***Abstract*—Audio captured in real-world environments is fundamentally limited by the finite dynamic range of analog-to-digital converters (ADCs). When high-dynamic-range features exceed this range, signals undergo destructive non-linearities such as clipping or wrap-around overflow (WAO), permanently degrading fidelity. This paper introduces USF-Audio, a clipping-free acquisition and reconstruction framework based on the Unlimited Sensing Framework (USF). By injecting a modulo non-linearity in the analog front-end, the proposed hardware preserves all amplitude information by folding large values into a bounded interval, preventing saturation or overload without sacrificing resolution. We develop a scalable reconstruction algorithm that inverts the modulo operation and also unifies recovery from classical clipping, WAO, and MSB loss. Hardware-validated experiments demonstrate that USF-Audio achieves substantial objective and perceptual improvements over state-of-the-art declipping methods, enabling high-dynamic-range, high-resolution audio capture with standard bit budgets.**

***Index Terms*—Audio signal processing, declipping, quantization, non-linear acquisition and unlimited sensing.**

I. Introduction

EUSIPCO 2025 featured a classical concert at the *Segesta Temple* in Palermo that offered an unexpectedly revealing audio experiment. Although the performance sounded impressive in person—but a handheld recording told a different story. On playback, the audio was marred by the familiar harsh, gritty, crackling artifacts of clipping: every gust of wind, every percussive transient, every swell of the orchestra pushed the device's analog-to-digital converter (ADC) beyond its limits, flattening peaks and shredding the natural dynamics. The scene was serene; the audio recording was not.

This episode underscores a broader reality. Despite the convenience of modern digital audio—its ability to *capture*, *store*, and *transmit* sound almost anywhere—acquisition systems remain *fundamentally constrained* by ADC's finite dynamic range. Once a signal exceeds the representable range, the digital system departs from linearity and applies a non-linear amplitude transformation.

More generally, this behavior can be viewed as an **overflow problem** [1]. In ADCs and digital pipelines, overflow manifests as (i) *clipping or saturation* (see Fig. 1 (a)), where large amplitudes are truncated [2], and (ii) *wrap-around overflow* or the effective loss of most-significant bits (MSBs) due to fixed-point constraints [3] (see Fig. 1 (b)). Although described using different terminology across communities, these effects share the same origin: amplitudes are forced into a bounded range through non-linear operations.

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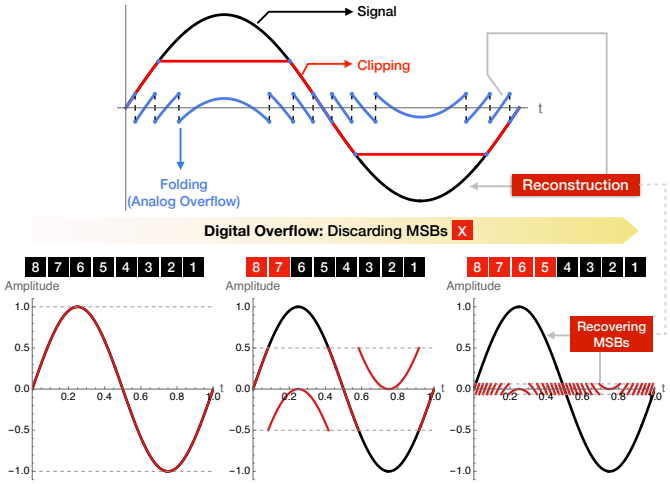


Fig. 1: Deliberately applying wrap-around overflow (WAO) eliminates the saturation or clipping overflow. Top: preventing signal clipping via analog-domain WAO (or modulo non-linearity, $\mathscr{M}$), bottom: recovering most-significant bits (MSBs) from least-significant bits (LSBs).

Clipping has therefore been widely studied in signal processing as an inverse problem, leading to extensive declipping and inpainting algorithms [4]–[6]. Wrap-around overflow, in contrast, has been treated primarily in hardware and computer arithmetic, where it is often regarded as a "catastrophic loss" of signal integrity [7]. In both settings, however, post-hoc remedies are inherently limited. Since overflow induces a permanent information loss, the measurements no longer uniquely encode the original signal, and recovery is approximate.

To address this limitation, we take a different perspective—*diamonds cut diamonds*. Rather than wrestling with samples corrupted by overflow, we re-think audio sensing so that the signal is encoded through a *deliberate, invertible non-linearity* rather than destructive, irreversible ones.

What's the new idea? Our work builds on the Unlimited Sensing Framework (USF), which advocates the joint design of hardware and algorithms [8]. In USF, a controlled modulo non-linearity ($\mathscr{M}$) is introduced in the analog domain prior to sampling and quantization. Unlike accidental overflow, the modulo operator, $\mathscr{M}$, *purposefully folds large amplitudes into a bounded interval* (see Fig. 1 (a)), ensuring

that high-dynamic-range (HDR) signals remain within the ADC's limits and thereby preventing destructive clipping. This redistribution of HDR amplitudes removes the conventional trade-off between dynamic range (DR) and digital resolution (DRes): for the same bit budget, USF enables simultaneous HDR capture and HDRes. Moreover, for sufficiently smooth signals, $\mathscr{M}$ admits stable inversion, allowing recovery of HDR waveforms with correspondingly HDRes.

Because the mathematical inverse problem depends solely on the measurement model—and not on whether folding occurs in analog circuitry or digital hardware—the **same reconstruction strategy applies to wrap-around overflow** and MSB truncation. In this way, a single framework can prevent clipping, recover from overflow, and reconstruct missing MSBs from LSBs, providing more reliable acquisition and potentially higher effective data rates.

Contribution. We introduce **USF-Audio**, a complete audio pipeline based on unlimited sampling, enabling saturation-free high-dynamic-range acquisition and principled reconstruction. Building on the USF measurement model, we develop a scalable algorithm that recovers unclipped audio and reconstructs signals even under wrap-around overflow. Our three main contributions are: (i) a hardware-realizable modulo front-end for HDR audio capture, (ii) an efficient reconstruction method tailored for large-scale audio signals, and iii) hardware-validated experiments showing substantial perceptual and objective gains over state-of-the-art declipping methods.

II. Recovery Guarantees for Overflow

Problem Statement. Let $g \in \mathcal{B}_\Omega$ be a HDR, continuous-time audio signal with maximum angular frequency Ω. Radically different from the conventional acquisition schemes, prior to sampling and quantization, g is folded in the analog domain via the non-linear modulo operation, defined as,

$$\mathscr{M}_\lambda : g \mapsto 2\lambda \left(\left[\!\!\left[\frac{g}{2\lambda} + \frac{1}{2} \right]\!\!\right] - \frac{1}{2} \right), \quad [\![g]\!] \overset{\text{def}}{=} g - \lfloor g \rfloor \quad \lambda > 0 \tag{1}$$

which yields bounded, low-dynamic-range signal, $y(t) = \mathscr{M}_\lambda(g(t))$. Uniform sampling of $y(t)$ leads to folded samples,

$$y[n] = \mathscr{M}_\lambda(g(nT)), \quad n \in \mathbb{I}_N \tag{2}$$

where $\mathbb{I}_N = \{0, 1, \cdots, N-1\}$. With $\{y[n]\}_{n\in\mathbb{I}_N}$ known, our goal is to reconstruct $\{g[n]\}_{n\in\mathbb{I}_N}$, where $g(t)$ can be recovered using Shannon interpolation formula.

Recovery Approach and Guarantees. Generally speaking, modulo signal recovery amounts to a combinatorial optimization: by definition, g can be decomposed as,

$$g[n] = y[n] + \varepsilon_g[n], \;\; \varepsilon_g[n] = 2\lambda \sum\nolimits_{m=0}^{M-1} \gamma_m u[n - n_m], \tag{3}$$

where $u(\cdot)$ is the unit-step function, and $\{\gamma_m \in \mathbb{Z}, n_m \in \mathbb{I}_N\}$ parametrize ε_g. Hence, the recovery problem can be posed as,

$$\min_{\varepsilon_g} \|\mathcal{R}(\widetilde{g}[n])\|_{\ell_p}, \;\; \text{s.t.} \;\; \begin{array}{l} \widetilde{g}[n] = y[n] + \varepsilon_g[n], \\ \varepsilon_g[n] \in 2\lambda\mathbb{Z}, \; p \geqslant 1 \end{array} \tag{4}$$

where $\mathcal{R}(\cdot)$ is a regularizer that promotes smoothness, *e.g.* high-order difference operator (see Theorem 1) [8]–[10] or linear prediction operator via Chebyshev polynomials [11].

The combinatorial optimization is messy and potentially intractable, since its computational complexity scales exponentially with both the DR $\|g\|_{\mathbf{L}_\infty}$ and sample size N. In the context of audio sensing, this is a particular issue due to: **(i) Voluminous data**: An audio clip with 20 sec duration and standard sampling frequency of 44-48 kHz naturally results in large-scale measurements; **(ii) HDR feature**: The audio waveforms exhibit HDR characteristics, due to (a) super-oscillation and (b) voiced-unvoiced transitions. Hence, recovering audio signals from folded samples via solving (4) are impractical.

To tackle this problem, the key insight in this paper is that, the solution space of (4) can be strictly bounded via the constrained smoothness prior of g. Under mild oversampling conditions, we demonstrate that a constructive solution can be derived by non-linearly filtering the folded samples y, enabling efficient, exact and unique audio recovery.

The main result is formalized in the following theorem:

Theorem 1. *Let $g \in \mathcal{B}_\Omega$ and the folded samples be $y[n] = \mathscr{M}_\lambda(g(nT)), n \in \mathbb{I}_N$. Then, $\{g[n]\}_{n\in\mathbb{I}_N}$ can be recovered up to an additive constant, provided that,*

$$T < \tfrac{1}{2\Omega} \quad \textit{and} \quad N \geqslant \left\lceil \tfrac{\ln\lambda - \ln\|g\|_{\mathbf{L}_\infty}}{\ln(\Omega T)} \right\rceil + \tfrac{6\|g\|_{\mathbf{L}_\infty}}{\lambda}. \tag{5}$$

Proof sketch. Let $\Delta^{(h)} g[n] = \Delta^{(h-1)} g[n+1] - \Delta^{(h-1)} g[n]$ be the h-th order finite difference of g.

Step 1 (Difference-Decay). We first show that, Δ^h which is high-pass, "annihilates" $g[n]$ (low-pass). Define

$$\mathcal{I}_T : L^2(\mathbb{R}) \to \ell^2(\mathbb{Z}), \;\; (\mathcal{I}_T x)[n] = \int_{nT}^{(n+1)T} x(t)\, dt, \;\; n \in \mathbb{Z}.$$

We start with $h = 1$: by definition, $\Delta^{(1)} g[n]$ is given by,

$$\left|\Delta^{(1)} g[n]\right| = |(\mathcal{I}_T \partial_t g)[n]| \leqslant T \|\partial_t g\|_{\mathbf{L}_\infty} \tag{6}$$

which follows from the mean-value theorem. Utilizing Bernstein's inequality [9], [10], we have, $\left|\Delta^{(1)} g[n]\right| \leqslant \Omega T \|g\|_{\mathbf{L}_\infty}$, yielding amplitude decay if $\Omega T < 1$. For $h = 2$, we can write,

$$\left|\Delta^{(2)} g[n]\right| \leqslant \Omega T \|f_1\|_{\mathbf{L}_\infty} \leqslant (\Omega T)^2 \|g\|_{\mathbf{L}_\infty}. \tag{7}$$

By definition, for finite difference depth h, we can derive,

$$\left|\Delta^{(h)} g[n]\right| = |(\mathcal{I}_T f_{h-1})[n]| \leqslant T \|\partial_t f_{h-1}\|_{\mathbf{L}_\infty} \tag{8}$$

where $f_{h-1}(t) = f_{h-2}(t+T) - f_{h-2}(t)$. Hence, by induction,

$$\|f_{h-1}\|_{\mathbf{L}_\infty} \leqslant \Omega T \|f_{h-2}\|_{\mathbf{L}_\infty} \to \|f_{h-1}\|_{\mathbf{L}_\infty} \leqslant (\Omega T)^{h-2} \|f_1\|_{\mathbf{L}_\infty}$$

and plugging (7) into the equation above leads to,

$$h \geqslant h_0 \overset{\text{def}}{=} \left\lceil \tfrac{\ln\lambda - \ln\|g\|_{\mathbf{L}_\infty}}{\ln(\Omega T)} \right\rceil \Longrightarrow \left|\Delta^{(h)} g[n]\right| \leqslant \lambda. \tag{9}$$

which essentially enables recovering $\Delta^{(h)} g$ from $\Delta^{(h)} y$.

Step 2 (Modulo Invariance at depth h). In view of (3), $\Delta^{(h)} y$

Wrap-around Overflow Audio Recovery Algorithm

Input: Wrap-around overflow samples $\{y[n]\}_{n=0}^{N-1}$.
1: Compute h_0 via (9).
2: Recover $\Delta^{(h_0)}g$ from $\Delta^{(h_0)}y$ via (11).
3: **for** $h = h_0$ to 2 **do**
4: Calculate $\mathcal{S}_2\left(\Delta^{(h)}\varepsilon_g[n]\right)$ via (12).
5: Retrieve β_h via (14).
6: **end for**
Output: The recovered audio signal $\widetilde{g}$.

can be decomposed as,

$$\Delta^{(h)}y[n] = \Delta^{(h)}g[n] - \Delta^{(h)}\varepsilon_g[n],\ \Delta^{(h)}\varepsilon_g[n] \in 2\lambda\mathbb{Z}. \quad (10)$$

Applying the modulo non-linearity to (10) results in,

$$\mathscr{M}_\lambda(\Delta^{(h_0)}y[n]) = \mathscr{M}_\lambda(\Delta^{(h_0)}g[n]) \overset{(9)}{=} \Delta^{(h_0)}g[n]. \quad (11)$$

Step 3 (Anti-differences). With $\Delta^{(h_0)}g[n]$ known, we can recover $g[n]$ up to a polynomial of degree-h. To find the coefficients, let $\mathcal{S}_h$ denote the h-th order anti-difference operator,

$$\mathcal{S}_2\left(\Delta^{(h)}\varepsilon_g[n]\right) = \Delta^{(h-2)}\varepsilon_g[n] + 2\lambda\beta_h n + 2\lambda\beta_{h-1} \quad (12)$$

where $\{\beta_h, \beta_{h-1}\} \in \mathbb{Z}^2, h \leqslant h_0$. Finding $\{\beta_h, \beta_{h-1}\}$ is equivalent to solve (4); yet solution approach is simplified due to smoothness, which can be evaluated as,

$$\begin{aligned}\mathcal{D}_2 &= \mathcal{S}_2\left(\Delta^{(h)}\varepsilon_g[n]\right) - \mathcal{S}_2\left(\Delta^{(h)}\varepsilon_g[n+J]\right)\\ &\overset{(10)}{=} (\Delta^{(h-2)}g[n] - \Delta^{(h-2)}g[n+J])\\ &\quad - (\Delta^{(h-2)}y[n] - \Delta^{(h-2)}y[n+J]) - 2\lambda\beta_h J\end{aligned}$$

and from triangular inequality and (9), we can derive that,

$$\left|\Delta^{(h-2)}g[n] - \Delta^{(h-2)}g[n+J]\right| \leqslant 2\|g\|_{\mathbf{L}_\infty} \leqslant 2^{h-1}\lambda\ . \quad (13)$$

In view of (9), we can write that, $2^{h_0-1}\lambda < \|g\|_{\mathbf{L}_\infty}$, and

$$\frac{1}{2\lambda J}\mathcal{D}_2 \in \frac{3\|g\|_{\mathbf{L}_\infty}}{\lambda J}\left[-\frac{1}{2}, \frac{1}{2}\right] - \beta_h, \quad h \leqslant h_0. \quad (14)$$

Hence, the integer β_h can be uniquely found from $\frac{1}{2\lambda J}\mathcal{D}_2$ if $J \geqslant \lceil 6\|g\|_{\mathbf{L}_\infty}/\lambda \rceil$ and $N \geqslant h_0 + J$ to guarantee sufficient sample size. Recursively, $\{\beta_h\}_{h=2}^{h_0}$ can be perfectly retrieved, which completes the proof of this theorem[1]. □

Theorem 1 provides exact recovery from modulo overflow samples with theoretical guarantees (see Algorithm 1), enabling reliable solution to retrieving MSBs from LSBs.

III. Practical and Robust Audio Recovery

Leveraging the commutativity between modulo non-linearity and high-order difference operator, a non-linear filtering approach—US-Alg, is developed based on Theorem 1. Despite its scalability, exactness and DR compressibility, a notable gap remains when dealing with analog folded measurements: the success of US-Alg relies on the on-grid assumption, $(g - y) \in 2\lambda\mathbb{Z}$. However, this condition is restrictive in real-world scenarios, due to hardware imperfection of modulo ADCs that yields non-ideal folding, $(g-y) \notin 2\lambda\mathbb{Z}$ [13], [14].

[1]Detailed proof and experimental description will be available in [12].

Robust USF-Audio Recovery Algorithm

Input: Noisy folded measurements $\{y[n]\}_{n=0}^{N-1}$.
1: **for** $l = 0$ to $L-1$ **do**
2: Compute $\{\underline{y}_l[n]\}_{n=0}^{N_l-2}$.
3: Compute $\{\widehat{\underline{y}}_l[k]\}_{k=K_l}^{N_l-2}$ via (16).
4: Solve (17) via the matrix pencil method [15].
5: Perform on-grid projection via (18).
6: Find $\{c_{m,l}\}_{m\in\mathbb{I}_{M_l}}$ via least-squares.
7: Reconstruct $\varepsilon_{g,l}$ and g_l via (15).
8: Concatenate $\{g_l\}$.
9: **end for**
Output: The recovered audio signal $\widetilde{g}$.

To tackle this issue and fundamentally eliminate clipping, we design a robust unfolding method that is agnostic to folding non-linearities. In literature, although several algorithmic approaches have been proposed for this purpose, such as Fourier-domain partitioning [16] and iterative method [10], [17], they suffer from scalability and numerical stability of audio data.

To this end, we adopt a *frame-wise processing strategy* prevalent in audio community [18]. Specifically, we unfold the samples locaally, in a frame-wise fashion. The key idea behind such local unfolding is to separate the smooth (g) and sparse (ε_g) components via an appropriate transform. In this context, the Fourier transform used in [16] is less suitable because signal segmentation inevitably induces spectral leakage.

Frame-wise DCT. In order to "sparsify" g locally, we employ the DCT (Discrete Cosine Transform) due to its energy compaction. Let y_l and $\underline{y}_l = y_l[n+1] - y_l[n]$ be the folded data of l-frame and its finite difference, then, (3) reduces to,

$$\underline{g}_l[n] = \underline{y}_l[n] + \underline{\varepsilon}_{g,l}[n],\ \ \underline{\varepsilon}_{g,l}[n] = \sum\nolimits_{m=0}^{M_l-1} c_{m,l}\delta[n - n_{m,l}], \quad (15)$$

where $c_{m,l} \in \mathbb{R}$, $n_{m,l} \in \mathbb{I}_{N_l}$ and N_l is the frame size. Performing DCT on the equation above leads to,

$$\begin{aligned}&\widehat{\underline{y}}_l[k] = \widehat{\underline{g}}_l[k] - \widehat{\underline{\varepsilon}}_{g,l}[k],\ \ \widehat{\underline{g}}_l[k] \approx 0, k \in \mathbb{I}_{N_l} \setminus \mathbb{I}_{K_l},\\ &\widehat{\underline{y}}_l[k] = \textstyle\sum_{n=0}^{N_l-2} \underline{y}_l[n]\cos\left(\frac{\pi}{N_l-1}\left(n+\frac{1}{2}\right)k\right)\end{aligned} \quad (16)$$

where $\{\widehat{\underline{y}}_l, \widehat{\underline{g}}_l, \widehat{\underline{\varepsilon}}_{g,l}\}$ is the DCT representation of $\{\underline{y}_l, \underline{g}_l, \underline{\varepsilon}_{g,l}\}$ and K_l is the bandwidth in the DCT domain. Consequently, the local unfolding problem can be posed as,

$$\min_{\{c_{m,l}, n_{m,l}\}} \sum_{k=K_l}^{N_l-2}\left|\widehat{\underline{y}}_l[k] + \sum\nolimits_{m=0}^{M_l-1} c_{m,l}e^{-\jmath\frac{2kn_{m,l}\pi}{N_l-1}}\right|^2 \quad (17)$$

which amounts to a standard high-resolution spectral estimation problem. For computational efficiency, we use the matrix pencil method [15]. On-grid projection, *i.e.*

$$n_{m,l} = -\left\lfloor\frac{(N_l-1)\omega_{m,l}}{2\pi} + \frac{1}{2}\right\rfloor, \quad m \in \mathbb{I}_{M_l} \quad (18)$$

are performed to enforce the integer spike locations, where $\{\omega_{m,l}\}$ are the estimated frequencies. With $\{n_{m,l}\}_{m\in\mathbb{I}_{M_l}}$

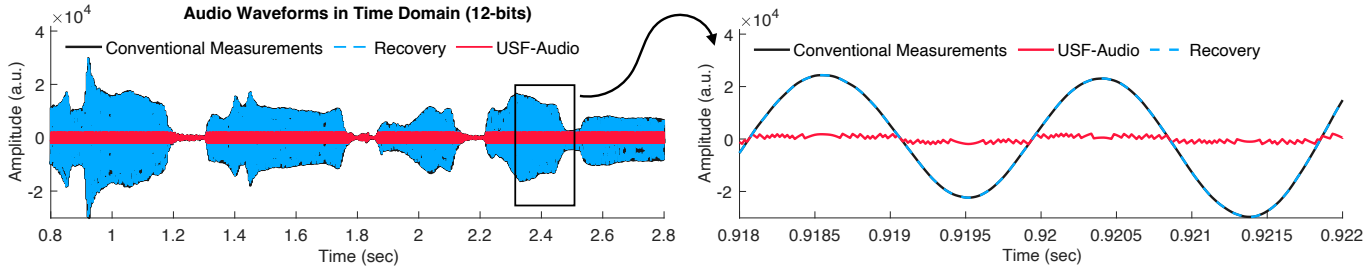


Fig. 2: Audio signal recovery from digital overflow. As shown in Fig. 1, dropping MSBs results in signal folding. However, these MSBs can be recovered by leveraging the smoothness of audio and speech data. Using Algorithm 1, we recover 4 MSBs from 12-bit measurements (LSBs) effectively recovering the 16-bit measurements.

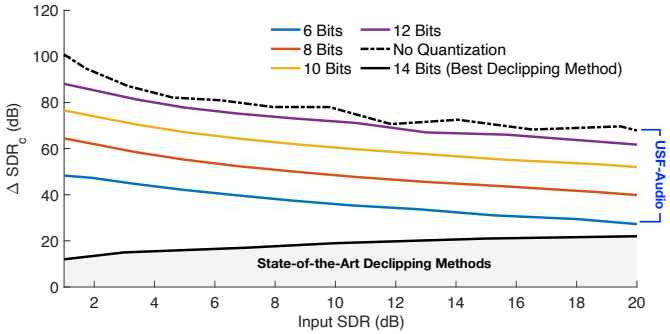


Fig. 3: USF-Audio vs state-of-the-art declipping methods [5], [6]. We quantify the performance of USF-Audio regarding with bits and clipping severity. Since clipping results in a permanent information loss, best declipping methods are capped at a 20 dB ceiling. Particularly, in severe clipping scenarios (input SDR: $[1, 3]$ dB), the improvement is at most 11 dB. In contrast, USF-Audio delivers a significant performance gain of > 32 dB in all cases.

known, the amplitudes could be retrieved via least-squares, and g could be recovered by concatenating the local reconstruction via $g_l \leftarrow y_l + \varepsilon_{g,l}$. The algorithm is stated in Algorithm 2.

IV. Hardware Experimental Validation

We experimentally validate USF-Audio in two complementary real-world scenarios:

1] recovery of lost MSBs under wrap-around overflow, and
2] elimination of audio clipping via modulo ADC hardware.

The key takeaway is as follows: USF-Audio *enables dynamic-range expansion (DRX) and distortion removal directly at the sensing stage, and the recovery guarantees of Theorem 1 translate into quantifiable hardware gains.*

Experiment 1: Recovering MSBs from LSBs (Wrap-Around Overflow). We demonstrate that bandlimited-prior combined with oversampling allows exact recovery of lost MSBs from LSB-only data, thereby expanding dynamic range without increasing nominal bit-depth. This experiment can alternatively be interpreted as *purposeful bit dropping* to enable higher data rates or reduced storage. **Protocol.** We record an 16-bit audio corpus (music, vocals, speech) using a mic and soundcard on a `MacBook Pro (2019)` with 60 kHz sampling rate. The recording spans the full microphone DR. Wrap-around overflow (WAO) is implemented by progressively removing MSBs, leaving only LSB information. We identify

TABLE I: Audio Declipping: Datasets, Experimental Parameters and Metrics.

Dataset	$\|g\|_\infty$	Digital Resolution		Noise Floor (dB)		λ_{clip}	N	Input SDR (dB)	ΔSDR (dB)	RMSE ($\times 10^{-2}$)
		ADC	USF-Audio	ADC	USF-Audio					
1 (Fig. 4)	6.63	0.20	0.02	41.45	23.39	0.60	2833230	2.71	33.16	7.39
2	5.32	0.20	0.0048	41.60	16.40	0.62	2908453	4.90	25.62	7.51
3	3.70	0.16	0.02	15.38	0.19	0.56	10000	2.72	32.55	6.54
4	6.63	0.20	0.02	40.97	23.18	0.59	2506310	2.35	32.10	7.56

the minimum bit-budget required for exact reconstruction. One recovered MSB corresponds to a $2\times$ dynamic-range expansion (DRX). **Results and Significance.** As shown in Fig. 2, our method (see Algorithm 1) retrieves 4 MSBs from 12 bits, offering $16\times$ DRX. This experiment validates Theorem 1: oversampling converts apparent information loss (modulo wrap or WAO) into recoverable structure. Practically, this means overflowed recordings once seen as a "catastrophic loss" [7] can be losslessly restored, and DR can be expanded without modifying the nominal quantizer resolution.

Experiment 2: Elimination of Audio Clipping (Hardware Modulo ADC). We demonstrate that analog-domain folding eliminates clipping and enables high-dynamic-range audio capture beyond conventional ADC limitations. **Protocol.** We capture audio from a `Beoplay P2` speaker using a custom-designed modulo ADC with $\lambda = 0.5$, sampling rate 250 kHz, and 6-bit quantization [16]. The corpus consists of $4\times$ clips (music, song, speech). A conventional ADC with identical digitization parameters records reference signals, serving as proxy ground truth. Clipping severity is quantified using input signal-to-distortion ratio (SDR) [5]: $\text{SDR}(g, g_{\text{clip}}) = 20 \log_{10} \frac{\|g\|_2}{\|g - g_{\text{clip}}\|_2}$. Severe clipping corresponds to $\lambda_{\text{clip}} \leq 0.1\|g\|_\infty$, yielding 1–3 dB SDR and implying more than 50% clipped samples. Recovery accuracy is measured by $\Delta\text{SDR} = \text{SDR}(g, \widetilde{g}) - \text{SDR}(g, g_{\text{clip}})$. **Results and Significance.** We first quantify the performance of USF-Audio pipeline under varying bit-budget and clipping severity. The bandlimited samples are generated by low-pass filtering white noise and quantized with distinct bits. Ground truth samples are artificially clipped at $\lambda_{\text{clip}} = \lambda$ to emulate conventional saturation. As shown in Fig. 3, our method consistently achieves a performance upgrade of at least 32 dB in all cases, whereas best declipping methods are capped at a performance ceiling, particularly in

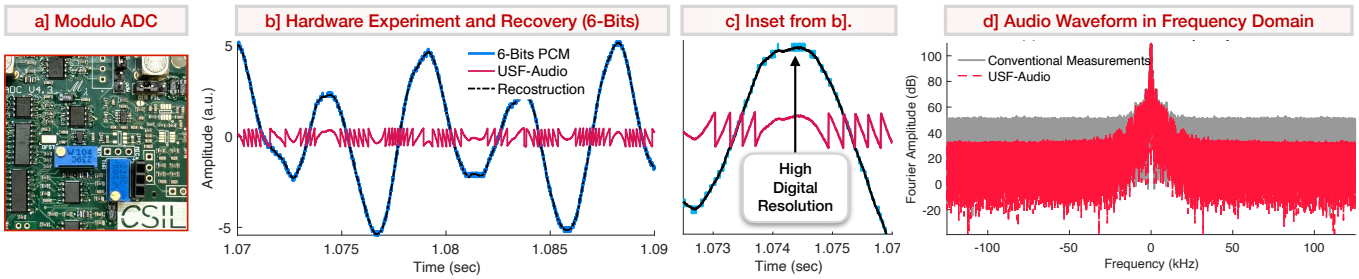


Fig. 4: End-to-end USF-Audio hardware pipeline. (a) Custom analog folding front-end. (b) Measured folded waveform and reconstructed signal versus conventional 6-bit PCM. (c) Inset showing enhanced digital resolution after unfolding. (d) Frequency-domain comparison of conventional and USF-Audio recovery. USF-Audio prevents clipping while improving effective resolution, achieving accurate recovery with 13.26-DRX and about 20 dB lower noise floor.

extreme clipping scenarios when input SDR: $[1, 3]$ dB.

Beyond numerical tests, we validate USF-Audio on hardware folded audio data. Despite large DRX, substantial folding activity, and hardware non-idealities, Algorithm 2 achieves accurate recovery across all cases. From Fig. 4, (i) The sensing DR is expanded while simultaneously improving measurement precision, with a $\approx 20\,\text{dB}$ noise-floor improvement (see Fig. 4 (b)). (ii) Under extreme clipping conditions where state-of-the-art declipping methods achieve at most $\approx 11\,\text{dB}$ ΔSDR [5], USF-Audio attains $\Delta\text{SDR} > 25\,\text{dB}$ (see Table I). This confirms that USF-Audio does not merely "repair" clipped audio; it fundamentally shifts the sensing bottleneck by replacing irreversible saturation with structured, invertible folding.

Overall Insight. Both experiments elucidate the principle: *overflow and clipping are not information-theoretic losses when folding is controlled and the signal is structured.* By co-designing hardware and algorithms, USF-Audio unlocks HDR audio capture with guarantees validated at hardware scale.

V. Conclusion

USF-Audio offers an alternative to conventional digital-audio by replacing destructive clipping with folding non-linearities, ensuring that high-dynamic-range signals are recoverable. Our hardware-validated experiments show that USF-Audio not only prevents saturation but also enables principled inversion of overflow or MSB loss, and wrap-around artifacts using a unified reconstruction model. Compared to state-of-the-art declipping methods, USF-Audio offers substantial improvements, demonstrating that dynamic-range limitations imposed by ADCs are not intrinsic but a consequence of traditional front-end design. We expect USF-based front-ends to enable compact, energy-efficient audio devices, adaptive dynamic-range expansion, and seamless integration with emerging neural codecs. We view USF-Audio as a first step toward future systems where analog non-linearity is not a failure mode but a design tool, enabling robust, high-fidelity sensing across increasingly demanding real-world environments.

References

[1] D. Goldberg, "What every computer scientist should know about floating-point arithmetic," *ACM Comput. Surv.*, vol. 23, no. 1, pp. 5–48, Mar. 1991.

[2] B. Defraene, T. van Waterschoot, H. J. Ferreau, M. Diehl, and M. Moonen, "Real-time perception-based clipping of audio signals using convex optimization," *IEEE Audio, Speech, Language Process.*, vol. 20, no. 10, pp. 2657–2671, Dec. 2012.

[3] J. Dattorro, "The implementation of recursive digital filters for high-fidelity audio," *Journal of the Audio Engineering Society*, no. 7-025, May 1989.

[4] J. Abel and J. Smith, "Restoring a clipped signal," in *IEEE Intl. Conf. on Acoustics, Speech, and Signal Processing*. IEEE, Apr. 1991.

[5] P. Zaviska, P. Rajmic, A. Ozerov, and L. Rencker, "A survey and an extensive evaluation of popular audio declipping methods," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 1, pp. 5–24, Jan. 2021.

[6] C. Gaultier, S. Kitić, R. Gribonval, and N. Bertin, "Sparsity-based audio declipping methods: Selected overview, new algorithms, and large-scale evaluation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 1174–1187, Feb. 2021.

[7] G. A. Constantinides, P. Y. K. Cheung, and W. Luk, "Synthesis of saturation arithmetic architectures," *ACM Trans. Des. Autom. Electron. Syst.*, vol. 8, no. 3, pp. 334–354, Jul. 2003.

[8] A. Bhandari, F. Krahmer, and R. Raskar, "On unlimited sampling," in *Intl. Conf. on Sampling Theory and Applications (SampTA)*, Jul. 2017.

[9] ——, "On unlimited sampling and reconstruction," *IEEE Trans. Sig. Proc.*, vol. 69, pp. 3827–3839, Dec. 2020.

[10] R. Guo and A. Bhandari, "Unlocking off-the-grid sparse recovery with unlimited sensing: Simultaneous super-resolution in time and amplitude," *IEEE J. Sel. Topics Signal Process.*, pp. 1–13, 2025.

[11] E. Romanov and O. Ordentlich, "Above the Nyquist rate, modulo folding does not hurt," *IEEE Signal Processing Letters*, vol. 26, no. 8, pp. 1167–1171, Aug. 2019.

[12] R. Guo, Y. Zhu, and A. Bhandari, "Re-thinking digital audio via unlimited sensing: Theory, algorithms, and hardware validation (under review)," *IEEE Audio, Speech, Language Process.*, 2026.

[13] D. Florescu and A. Bhandari, "Unlimited sampling with local averages," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022, pp. 5742–5746.

[14] A. Bhandari, "Back in the US-SR: Unlimited sampling and sparse super-resolution with its hardware validation," *IEEE Signal Process. Lett.*, vol. 29, pp. 1047–1051, Mar. 2022.

[15] Y. Hua and T. Sarkar, "Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 5, pp. 814–824, May 1990.

[16] A. Bhandari, F. Krahmer, and T. Poskitt, "Unlimited sampling from theory to practice: Fourier-Prony recovery and prototype ADC," *IEEE Trans. Sig. Proc.*, pp. 1131–1141, Sep. 2021.

[17] R. Guo and A. Bhandari, "ITER-SIS: Robust unlimited sampling via iterative signal sieving," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, Jun. 2023.

[18] A. Adler, V. Emiya, M. G. Jafari, M. Elad, R. Gribonval, and M. D. Plumbley, "Audio inpainting," *IEEE Audio, Speech, Language Process.*, vol. 20, no. 3, pp. 922–932, Mar. 2012.